

BINOMIAL COEFFICIENTS

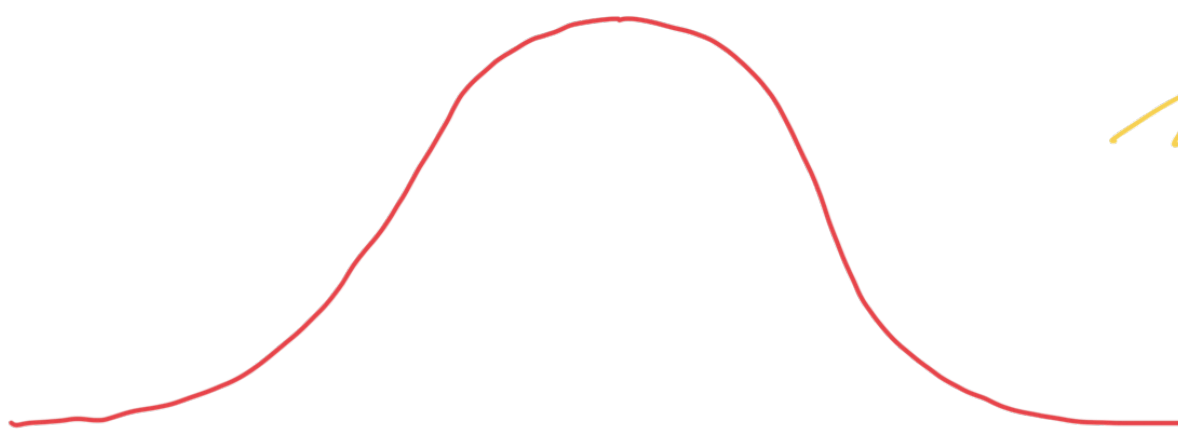
UNORDERED SAMPLES r OUT OF n WITHOUT REPLACEMENT

$$\text{IN } \binom{n}{r} = \frac{n!}{(n-r)!r!}$$

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$$

$(x+y)^0$					1	
$(x+y)^1$				1	1	
$(x+y)^2$			1	2	1	
$(x+y)^3$		1	3	3	1	
$(x+y)^4$	1	4	6	4	1	
$(x+y)^5$	1	5	10	10	5	1

$\binom{n}{r}$ related to N distribution



MULTINOMIAL DISTRIBUTION

n objects divided into r classes of size $n_1, n_2, \dots, n_r$ (order not important)

$$\binom{n}{n_1, n_2, \dots, n_r} = \frac{n!}{n_1! n_2! \dots n_r!}$$

⚡
not product

assumes $(n_1 + n_2 + \dots + n_r)$

Proof:

$$\binom{n}{n_1, n_2, \dots, n_r} = \binom{n}{n_1} \binom{n-n_1}{n_2} \binom{n-n_1-n_2}{n_3} \dots \binom{n-n_1-\dots-n_{r-1}}{n_r}$$

$$= \frac{n!}{n_1! \cancel{(n-n_1)!}} \cdot \frac{\cancel{(n-n_1)!}}{n_2! \cancel{(n-n_1-n_2)!}} \cdot \dots \cdot \frac{\cancel{(n-n_1-\dots-n_{r-1})!}}{n_r! (0!)}$$